

NB+C Project Number:

100869

Client:

Network Building + Consulting, LLC

Carrier:

Verizon

Carrier Site Name:

ALL WILLIAMS RELO

Carrier Site Number:

16824962

Carrier Project:

TRADITIONAL Capacity



1777 Sentry Parkway W, Veva 17, Suite 400, Blue Bell, PA 19422

Structural Mount Analysis

Code Parameters

Criteria and Parameters	
Codes and Standards	
International Building Code	2021 International Building Code
ASCE Standard	ASCE 7-16
TIA Standard	ANSI/TIA-222-H
Design Methodology	TIA MA
Wind Loading Hazard Data	
Risk Category	II
Wind Speed, V	113 mph
Maintenance Wind Speed, V _m	30 mph
Exposure Category	C
Ground Elevation, z _s	260.84 ft
Topographic Method	Method 1
Topographic Factor, K _{zt}	1
Ice Loading Hazard Data	
Ice Wind Speed, V _{ice}	40 mph
Ice Thickness, t _i	1 in
Seismic Loading Hazard Data	
0.2-Second Period, S _s	0.168 g
1-Second Period, S ₁	0.047 g
Site Class	D - Default (see Section 11.4.3)
Maximum 0.2-Second Period, S _{MS}	0.269 g
Maximum 1-Second Period, S _{M1}	0.113 g
Design 0.2-Second Period, S _{DS}	0.179 g
Design 1-Second Period, S _{D1}	0.075 g
Maintenance Loading & Additional Loading	
Maintenance Load, L _m	500 lbf
Maintenance Load, L _v	250 lbf
Area Loading Dead	5 psf
Area Loading Ice	10 psf
Annex S Load Modification Factors	
Load Modification Factor Wind, K _{es}	1.00
Load Modification Factor Ice, K _{es}	1.00

Structure Data

Structure Information	
Supported Structure	
Centerline	148 ft
Type	Platform
Number of Sectors	3
Number of Sectors Modeled	3
Condition	Proposed
Mount Reference Node Label	N1
Supporting Structure	
Height	150 ft
Type	Monopole
Structure on Supporting Structure?	No

Forces Summary

Forces Summary	
Wind Pressure and Maintenance Wind Pressure	
Wind Pressure, q _z	42.28 psf
Maintenance Wind Pressure, q _{mz}	2.98 psf
Ice Wind Pressure and Escalated Ice Thickness	
Ice Wind Pressure, q _{iz}	5.30 psf
Escalated Ice Thickness, t _{iz}	1.16 in
Seismic Effects	
Vertical Seismic Effect, E _v	0.036
Horizontal Seismic Effect, E _h	0.090

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Structural Mount Analysis - Calculations Summary

Analysis Criteria

Factors and Calculations	
General	
Rooftop Wind Speed-Up Factor, K_s	1
Wind Direction Probability, K_d	0.95
Ground Elevation Factor, K_e	0.991
Gust Factor, G_H	1
Shielding Factor, K_a	0.9
Wind Calculations	
Velocity Pressure, K_z	1.375
Importance Factor, I_{wind}	N/A
Reduced Wind Pressure, $q_z * K_a * K_{es}$	38.055
Ice Calculations	
Height Escalation Factor, K_{iz}	1.162
Importance Factor, I_{ice}	1
Ice Wind Pressure, $q_{iz} * K_a * K_{es}$	4.768
Escalated Ice Thickness, $t_{iz} * K_{es}$	1.162
Seismic Calculations	
Importance Factor, $I_{earthquake}$	1
Reduction Factor, p	1
Response Modification Factor, R	2
Seismic Response Coefficient, C_s	0.090
C_s *Amplification Factor	0.108
Amplification Factor, A_s	1

RISA-3D Model Information

RISA-3D Information	
Information per Imported RISA-3D Model	
Unit Length	in
Unit Force	lbf
Unit Linear Force	plf
Unit Moment	lbf-in
Unit Area Load	ksf
Model Axes	XYZ
Steel Design Code	AISC 15th (360-16): LRFD
RISA-3D Version	23.00003
Filepath	C:\~Development\MAT\713949-VZW_MT_LO_H.r3d

Structure EPA & Weight

EPA & Weight Summary			
Ice Thickness	0.0 Inch	0.5 Inch	1.0 Inch
Front	59.52 ft²	81.65 ft²	104.91 ft²
Side	59.52 ft²	81.65 ft²	104.91 ft²
Weight	1,449.89 lbf	2,173.33 lbf	3,144.30 lbf

* Structure EPAs do not include shielding factors in calculations

** Structure EPAs & weights should only be used in lieu of manufacturer or more accurate data

*** Structure EPAs & weights only include members modeled in RISA-3D.

Results - Envelope Solution

Structure Capacity = 25.1 % - Pass

Capacity Summary						
Section Set	Shape	Bending Critical Member	Bending Capacity %	Shear Critical Member	Shear Capacity %	Pass / Fail
Face Horizontal	HSS3X3X3	M1	13.8%	M1	4.2%	Pass
Stand Off Arm	HSS6X3X3	M20	13.9%	M12	10.9%	Pass
Plan Brace	HSS3X3X3	M22	11.3%	M13	2.4%	Pass
Grating support Angle	L2X2X3	M18	24.8%	M10	3.4%	Pass
Kicker	L2X2X4	M104	17.2%	M95	0.3%	Pass
Bottom End connection	HSS3X3X4	M15	21.4%	M15	7.3%	Pass
Mount Pipes	PIPE_2.5	A-2	25.1%	A-3	4.3%	Pass
Support Rail	PIPE_1.5	M31	19.5%	M31	6.7%	Pass
Top End Connection	PL3/8X4	M37	14.7%	M37	3.9%	Pass
Weldment	PIPE_3.0	M88	12.8%	M88	3.9%	Pass

Structural Mount Analysis - Loading Summary

Appurtenance CaAa and Wind Loads																
Appurtenance	0°	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	Area Calculation	Wind Press.	Weight	
NNH-65C-R2B	11.39 ft²	10.46 ft²	8.59 ft²	7.66 ft²	8.59 ft²	10.46 ft²	11.39 ft²	10.46 ft²	8.59 ft²	7.66 ft²	8.59 ft²	10.46 ft²	TIA	42.28 psf	51.60 lbf	
NNH-65C-R2B	11.39 ft²	10.46 ft²	8.59 ft²	7.66 ft²	8.59 ft²	10.46 ft²	11.39 ft²	10.46 ft²	8.59 ft²	7.66 ft²	8.59 ft²	10.46 ft²	TIA	42.28 psf	51.60 lbf	
XXDWMM-12.5-65-8T / RT4401-48A	1.54 ft²	1.34 ft²	0.95 ft²	0.75 ft²	0.95 ft²	1.34 ft²	1.54 ft²	1.34 ft²	0.95 ft²	0.75 ft²	0.95 ft²	1.34 ft²	TIA	42.28 psf	23.20 lbf	
RF4440d-13A	1.87 ft²	1.68 ft²	1.31 ft²	1.13 ft²	1.31 ft²	1.68 ft²	1.87 ft²	1.68 ft²	1.31 ft²	1.13 ft²	1.31 ft²	1.68 ft²	TIA	42.28 psf	72.50 lbf	
RF4439d-25A	1.87 ft²	1.71 ft²	1.40 ft²	1.25 ft²	1.40 ft²	1.71 ft²	1.87 ft²	1.71 ft²	1.40 ft²	1.25 ft²	1.40 ft²	1.71 ft²	TIA	42.28 psf	74.70 lbf	
RxxDC-6627-PF-48	4.06 ft²	3.82 ft²	3.33 ft²	3.09 ft²	3.33 ft²	3.82 ft²	4.06 ft²	3.82 ft²	3.33 ft²	3.09 ft²	3.33 ft²	3.82 ft²	TIA	42.28 psf	32.00 lbf	
MT6407-77A	4.70 ft²	3.99 ft²	2.56 ft²	1.84 ft²	2.56 ft²	3.99 ft²	4.70 ft²	3.99 ft²	2.56 ft²	1.84 ft²	2.56 ft²	3.99 ft²	TIA	42.28 psf	81.50 lbf	

Appurtenance Ice CaAa and Ice Wind Loads															
Appurtenance	0°	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	Area Calculation	Ice Wind Press.	Weight
NNH-65C-R2B	13.56 ft²	12.61 ft²	10.70 ft²	9.74 ft²	10.70 ft²	12.61 ft²	13.56 ft²	12.61 ft²	10.70 ft²	9.74 ft²	10.70 ft²	12.61 ft²	TIA	5.30 psf	215.86 lbf
NNH-65C-R2B	13.56 ft²	12.61 ft²	10.70 ft²	9.74 ft²	10.70 ft²	12.61 ft²	13.56 ft²	12.61 ft²	10.70 ft²	9.74 ft²	10.70 ft²	12.61 ft²	TIA	5.30 psf	215.86 lbf
XXDWMM-12.5-65-8T / RT4401-48A	2.12 ft²	1.89 ft²	1.44 ft²	1.21 ft²	1.44 ft²	1.89 ft²	2.12 ft²	1.89 ft²	1.44 ft²	1.21 ft²	1.44 ft²	1.89 ft²	TIA	5.30 psf	54.74 lbf
RF4440d-13A	2.49 ft²	2.28 ft²	1.85 ft²	1.64 ft²	1.85 ft²	2.28 ft²	2.49 ft²	2.28 ft²	1.85 ft²	1.64 ft²	1.85 ft²	2.28 ft²	TIA	5.30 psf	117.00 lbf
RF4439d-25A	2.49 ft²	2.31 ft²	1.96 ft²	1.78 ft²	1.96 ft²	2.31 ft²	2.49 ft²	2.31 ft²	1.96 ft²	1.78 ft²	1.96 ft²	2.31 ft²	TIA	5.30 psf	121.58 lbf
RxxDC-6627-PF-48	5.00 ft²	4.74 ft²	4.21 ft²	3.95 ft²	4.21 ft²	4.74 ft²	5.00 ft²	4.74 ft²	4.21 ft²	3.95 ft²	4.21 ft²	4.74 ft²	TIA	5.30 psf	122.93 lbf
MT6407-77A	5.74 ft²	4.96 ft²	3.42 ft²	2.65 ft²	3.42 ft²	4.96 ft²	5.74 ft²	4.96 ft²	3.42 ft²	2.65 ft²	3.42 ft²	4.96 ft²	TIA	5.30 psf	155.54 lbf

